Regression Analysis and CAPM - MIT 18.642

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1 Lecture 6 - MIT 18.642- Linear Regression Modelling for the Capital Asset Pricing Model (CAPM)

This code is based on:

- fm_casestudy_0_InstallOrLoadLibraries.r
- fm_casestudy_1_0.r
- fm_casestudy_1_rcode.r

2 Capital Asset Pricing Model Theory

Sharpe (1964) and Lintner (1965) developed the Capital Asset Pricing Model for a market in which investors have the same expectations, hold portfolios of risky assets that are mean-variance efficient, and can borrow and lend money freely at the same risk-free rate. In such a market, the expected return of asset j is

$$E[Rj] = R_{riskfree} + \beta_j (E[R_{Market}] - R_{riskfree})$$

$$\beta_j = Cov[R_j, R_{Market}]/Var[R_{Market}]$$

where R_{Market} is the return on the market portfolio and $R_{riskfree}$ is the return on the risk-free asset.

Consider fitting the simple linear regression model of a stock's daily excess return on the market-portfolio daily excess return, using the S&P 500 Index as the proxy for the market return and the 3-month Treasury constant maturity rate as the risk-free rate. The linear model is given by:

$R^*_{j,t} = \alpha_j + \beta_j R_{Market,t} + \varepsilon_{j,t}$, t=1,2, … .

where $\varepsilon_{j,t}$ are white noise: $WN(0, \sigma^2)$.

Under the assumptions of the CAPM, the regression parameters (α_j, β_j) are such that β_j is the same as in the CAPM model, and α_j is zero.

3 Install and Load Packages

```r
# code based on fm_casestudy_0_InstallOrLoadLibraries.r

# Install/load R packages
# Collect historical financial data from internet
# Create time series data matrix: casestudy1.data0.0
#   Closing prices on stocks (BAC, GE, JDSU, XOM)
#   Closing values of indexes (SP500)
#   Yields on constant maturity US rates/bonds (3MO, 1YR, 5YR, 10 YR)
#   Closing price on crude oil spot price



# 0. Install and load packages ----
#
# 0.1 Install packages ---
# Load packages into R session (and install if necessary)

# require() returns TRUE if the package is successfully loaded
# and FALSE if it is not found.
if (!require("pacman")) install.packages("pacman")
if (!require("conflicted")){
  # Install specific version from GitHub if not already present
  devtools::install_github("r-lib/conflicted")
}

# Note: p_load checks for existence before attempting installation
pacman::p_load(
  quantmod,  # Financial quantitative modeling
  tseries,   # Time series analysis
  vars,      # Vector Autoregressive models
  fxregime,  # Exchange rate regimes
  moments,
  rugarch,
  xts,
  graphics,
  tidyquant,
  tidyverse,
  fpp2
  )
```

```r
# 0.2 Load packages into R session
```

```r
library("quantmod")
library("tseries")
library("vars")
library("fxregime")
library("moments")
library("rugarch")
library("graphics")
```

4 Load Data Set

```r
# code based on fm_casestudy_1_0.r

# 1. Load data into R session ----

# 1.1  Stock Price Data from Yahoo
#   Apply quantmod(sub-package TTR)  function
#     getYahoodata
#         Returns historical data for any symbol at the website
#         http://finance.yahoo.com

# 1.1.1 Set start and end date for collection in YYYYMMDD (numeric) format

date.start <-as.Date("2000-01-01")
date.end <-as.Date("2013-05-31")

# 1.1.2 Collect historical data for S&P 500 Index

SP500<-getSymbols("^GSPC",
                  src= "yahoo",
                  from= date.start,
                  to=date.end,
                  auto.assign= FALSE
                  )
chartSeries(SP500[,1:5])
```


```
#1.1.3 Collecthistoricaldatafor4stocks
GE <- getSymbols("GE",
                 src= "yahoo",
                 from= date.start,
                 to=date.end,
                 auto.assign= FALSE)
BAC <- getSymbols("BAC",
                  src= "yahoo",
                  from= date.start,
                  to= date.end,
                  auto.assign= FALSE)
XOM <- getSymbols("XOM",
                  src= "yahoo",
                  from= date.start,
                  to= date.end,
                  auto.assign= FALSE)
JDSU <- getSymbols("VIAV",
                   src="yahoo",
                   from= date.start,
                   to = date.end,
                   auto.assign = FALSE)
```

```r
# 1.1.4 Details of data object GE from getYahoodata

# GE is a matrix object with
# row dimension equal to the number of dates
#     column dimension equal to 9
is.matrix(GE)
```

```
[1] TRUE
```

```r
dim(GE)
```

```
[1] 3372    6
```

```r
# Print out the first and last parts of the matrix:
head(GE,2)
```

```
            GE.Open  GE.High   GE.Low GE.Close GE.Volume GE.Adjusted
2000-01-03 244.4143 245.5126 238.3239 239.6219   4605131    129.4379
2000-01-04 235.2288 236.4269 230.0370 230.0370   4615898    124.2604
```

```r
tail(GE,2)
```

```
            GE.Open  GE.High   GE.Low GE.Close GE.Volume GE.Adjusted
2013-05-29 112.2389 113.5808 112.1430 113.2932   8187170    91.14697
2013-05-30 113.0057 113.8683 112.6223 113.1015   6123627    90.99277
```

```r
# Some attributes of the object GE

mode(GE)  # storage mode of GE is "numeric"
```

```
[1] "numeric"
```

```r
class(GE) # object-oriented class(es) of GE are "xts" and "zoo"
```

```
[1] "xts" "zoo"
```

```r
# xts is an extensible time-series object from the package xts

# zoo is an object storing ordered observations with an index attribute
# Important zoo functions
#   coredata() extracts or replaces core data
#   index() extracts or replaces the  (sort.by) index of the object
```

```r
# 1.2 Federal Reserve Economic Data (FRED) from the St. Louis Federal Reserve
#  Apply quantmod  function
#      getSymbols( seriesname, src="FRED")
#
#      Returns historical data for any symbol at the website
#      http://research.stlouisfed.org/fred2/

# Series name | Description

# DGS3MO      | 3-Month Treasury, constant maturity rate
# DGS1        | 1-Year Treasury, constant maturity rate
# DGS5        | 5-Year Treasury, constant maturity rate
# DGS10       | 10-Year Treasury, constant maturity rate

# DAAA        | Moody's Seasoned Aaa Corporate Bond Yield
# DBAA        | Moody's Seasoned Baa Corporate Bond Yield

# DCOILWTICO  | Crude Oil Prices: West Text Intermediate (WTI) - Cushing, Oklahoma

# 1.2.1 Default setting collects entire series
#       and assigns to object of same name as the series
getSymbols("DGS3MO", src="FRED")
```

```
[1] "DGS3MO"
```

```r
getSymbols("DGS1", src="FRED")
```

```
[1] "DGS1"
```

```r
getSymbols("DGS5", src="FRED")
```

```
[1] "DGS5"
```

```r
getSymbols("DGS10", src="FRED")
```

```
[1] "DGS10"
```

```r
getSymbols("DAAA", src="FRED")
```

```
[1] "DAAA"
```

```r
getSymbols("DBAA", src="FRED")
```

```
[1] "DBAA"
```

```r
getSymbols("DCOILWTICO", src="FRED")
```

```
[1] "DCOILWTICO"
```

```r
# Each object is a 1-column matrix with time series data
#   The column-name is the same as the object name
is.matrix(DGS3MO) #
```

```
[1] TRUE
```

```r
dim(DGS3MO)
```

```
[1] 11660     1
```

```r
head(DGS3MO)
```

```
           DGS3MO
1981-09-01  17.01
1981-09-02  16.65
1981-09-03  16.96
1981-09-04  16.64
1981-09-07     NA
1981-09-08  16.54
```

```r
tail(DGS3MO)
```

```
           DGS3MO
2026-05-04   3.70
2026-05-05   3.69
2026-05-06   3.69
2026-05-07   3.69
2026-05-08   3.69
2026-05-11   3.70
```

```r
mode(DGS3MO)
```

```
[1] "numeric"
```

```r
class(DGS3MO)
```

```
[1] "xts" "zoo"
```

```r
# 2.0   Merge data series together

# 2.1 Create data frame with all FRED series
#
#   Useful functions/methods for zoo objects
#     merge()
#     lag (lag.zoo)
#     diff()
#     window.zoo()
#     na.locf() # replace NAs by last previou non-NA
#     rollmean(), rollmax() # compute rolling functions, column-wise

fred.data0<-merge(
  DGS3MO,
  DGS1,
  DGS5,
  DGS10,
  DAAA,
  DBAA,
  DCOILWTICO)["2000::2013-05"]

# Determine data dimensions
dim(fred.data0)
```

```
[1] 3500    7
```

```r
class(fred.data0)
```

```
[1] "xts" "zoo"
```

```r
# Check first and last rows in object
head(fred.data0,2)
```

```
           DGS3MO DGS1 DGS5 DGS10 DAAA DBAA DCOILWTICO
2000-01-03   5.48 6.09  6.5  6.58 7.75 8.27         NA
2000-01-04   5.43 6.00  6.4  6.49 7.69 8.21      25.56
```

```r
tail(fred.data0,2)
```

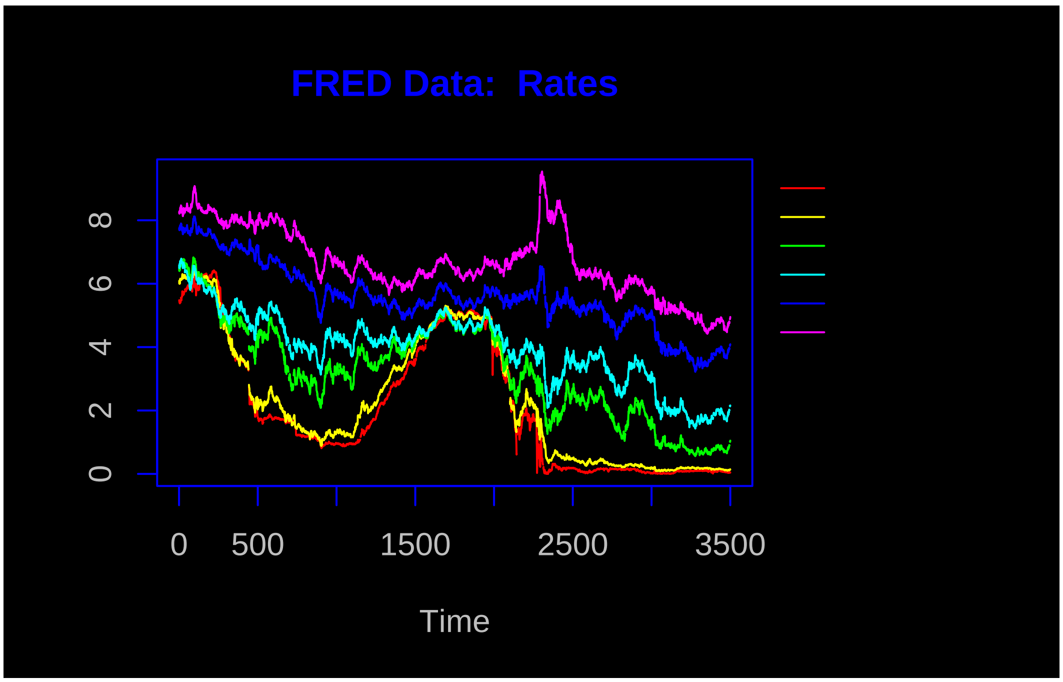
```
           DGS3MO DGS1 DGS5 DGS10 DAAA DBAA DCOILWTICO
2013-05-30   0.04 0.13 1.01  2.13 4.06 4.90      93.57
2013-05-31   0.04 0.14 1.05  2.16 4.09 4.95      91.93
```

```r
# Count the number of NAs in each column
apply(is.na(fred.data0),2,sum)
```

```
    DGS3MO       DGS1       DGS5      DGS10       DAAA       DBAA DCOILWTICO
       144        144        144        144        144        144        134
```

```r
# Plot the rates series all togehter
opar <- par(c("fg", "bg", "col.axis", "col.lab", "col.main", "col.sub"))

par(fg="blue",bg="black",
    col.axis="gray",
    col.lab="gray",
    col.main="blue",
    col.sub="blue",
    mar = c(5, 4, 4, 8),   # extra space on the right
    xpd = TRUE)            # allow drawing outside plot


ts.plot(as.ts(fred.data0[,1:6]),
        col=rainbow(6),
```

```
         bg="black",
         main="FRED Data:  Rates")
par(opar)

# add legend to plot
graphics::legend("topright",
        inset = c(-0.35, 0),    # Legend outside
        legend=dimnames(fred.data0)[[2]][1:6],
        lty=rep(1,times=6),
        col=rainbow(6),
        cex=0.75)
```




```
# Plot the Crude Oil PRice
chartSeries(to.monthly(fred.data0[,"DCOILWTICO"]),
            main="FRED Data: Crude Oil (WTI)")
```


```r
chartSeries(to.monthly(XOM[,1:5]))
```

```r
# 2.2 Merge the closing prices for the stock market data series
yahoo.data0 <- merge(
  BAC[, "BAC.Close"],
  GE[, "GE.Close"],
  JDSU[, "VIAV.Close"],
  XOM[, "XOM.Close"],
  SP500[, "GSPC.Close"]
)

dimnames(yahoo.data0)[[2]]<-c("BAC","GE","JDSU","XOM","SP500")
yahoo.data0.0<-zoo(x=coredata(yahoo.data0),
                   order.by=as.Date(time(yahoo.data0)))

# fred.data0 is already indexed by this scale
```

```r
# 2.3 Merge the yahoo and Fred data together

# 2.3.1 merge with all dates
casestudy1.data0 <- merge(yahoo.data0.0, fred.data0)

dim(casestudy1.data0)
```

```
[1] 3500   12
```

```r
apply(is.na(casestudy1.data0),2,sum)
```

```
       BAC         GE       JDSU        XOM      SP500     DGS3MO       DGS1 
       128        128        128        128        128        144        144 
      DGS5      DGS10       DAAA       DBAA DCOILWTICO 
       144        144        144        144        134 
```

```r
# 2.3.2 Subset out days when SP500 is not missing (not == NA)

index.notNA.SP500 <- which(is.na(coredata(casestudy1.data0$SP500))==FALSE)
casestudy1.data0.0 <- casestudy1.data0[index.notNA.SP500,]

head(casestudy1.data0.0,2)
```

```
                BAC       GE     JDSU      XOM   SP500 DGS3MO DGS1 DGS5 DGS10
2000-01-03 24.21875 239.6219 427.7588 39.15625 1455.22   5.48 6.09  6.5  6.58
```

```
2000-01-04 22.78125 230.0370 389.3629 38.40625 1399.42   5.43 6.00  6.4  6.49
           DAAA DBAA DCOILWTICO
2000-01-03 7.75 8.27         NA
2000-01-04 7.69 8.21      25.56
```

```r
tail(casestudy1.data0.0,2)
```

```
             BAC       GE     JDSU   XOM   SP500 DGS3MO DGS1 DGS5 DGS10 DAAA
2013-05-29 13.48 113.2932 7.713311 92.08 1648.36   0.05 0.14 1.02  2.13 4.04
2013-05-30 13.83 113.1015 7.810011 92.09 1654.41   0.04 0.13 1.01  2.13 4.06
           DBAA DCOILWTICO
2013-05-29 4.88      93.13
2013-05-30 4.90      93.57
```

```r
apply(is.na(casestudy1.data0.0)==TRUE, 2,sum)
```

```
       BAC         GE       JDSU        XOM      SP500     DGS3MO       DGS1
         0          0          0          0          0         24         24
      DGS5      DGS10       DAAA       DBAA DCOILWTICO
        24         24         24         24         14
```

```r
# Remaining missing values are for interest rates and the crude oil spot price
# There are days when the stock market is open
# but the bond market and/or commodities market is closed
# For the rates and commodity data, replace NAs with previoius non-NA values
casestudy1.data0.00 <- na.locf(casestudy1.data0.0)

apply(is.na(casestudy1.data0.00),2,sum)
```

```
       BAC         GE       JDSU        XOM      SP500     DGS3MO       DGS1
         0          0          0          0          0          0          0
      DGS5      DGS10       DAAA       DBAA DCOILWTICO
         0          0          0          0          1
```

```r
# Only 1 NA left, the first DCOILWTICO value
```

5 Save Data Set

```r
save(
  file="mit18_642_f24_lec06_1_3 - CAPM Case Study.RData",
  list=ls()
  )
```

6 Historical Financial Data

```r
load("mit18_642_f24_lec06_1_3 - CAPM Case Study.RData")
```

```r
# code based on fm_casestudy_1_rcode.r

dim(casestudy1.data0.00)
```

```
[1] 3372   12
```

```r
names(casestudy1.data0.00)
```

```
 [1] "BAC"         "GE"          "JDSU"        "XOM"         "SP500"
 [6] "DGS3MO"      "DGS1"        "DGS5"        "DGS10"       "DAAA"
[11] "DBAA"        "DCOILWTICO"
```

```r
head(casestudy1.data0.00,2)
```

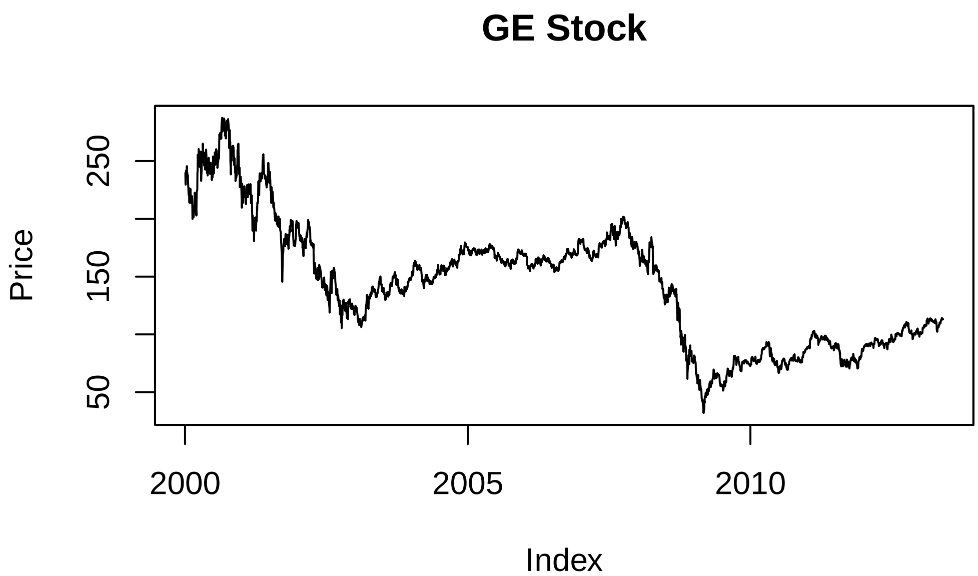
```
                BAC       GE     JDSU      XOM   SP500 DGS3MO DGS1 DGS5 DGS10
2000-01-03 24.21875 239.6219 427.7588 39.15625 1455.22   5.48 6.09  6.5  6.58
2000-01-04 22.78125 230.0370 389.3629 38.40625 1399.42   5.43 6.00  6.4  6.49
           DAAA DBAA DCOILWTICO
2000-01-03 7.75 8.27         NA
2000-01-04 7.69 8.21      25.56
```

```r
tail(casestudy1.data0.00,2)
```

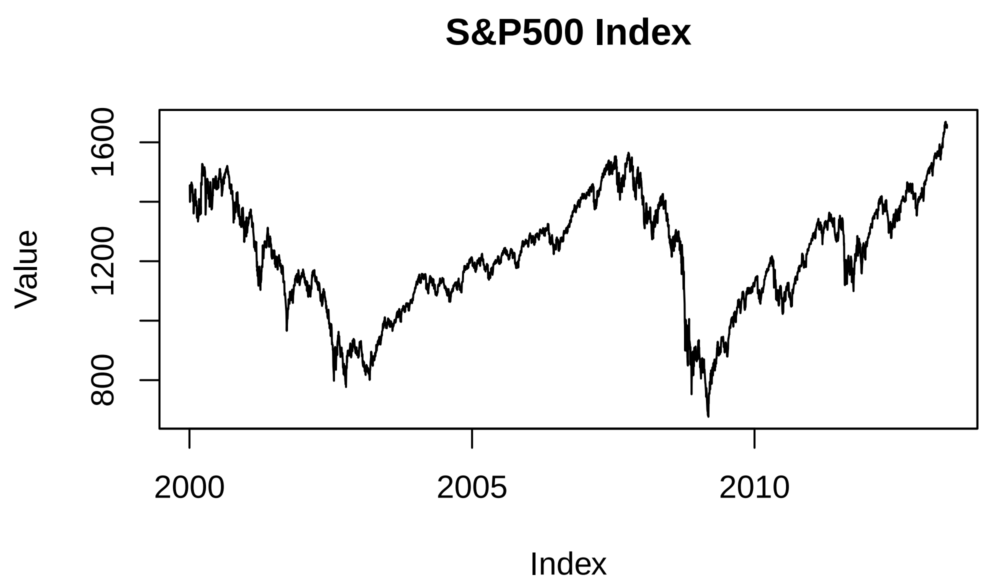
```
             BAC       GE     JDSU   XOM   SP500 DGS3MO DGS1 DGS5 DGS10 DAAA
2013-05-29 13.48 113.2932 7.713311 92.08 1648.36   0.05 0.14 1.02  2.13 4.04
2013-05-30 13.83 113.1015 7.810011 92.09 1654.41   0.04 0.13 1.01  2.13 4.06
           DBAA DCOILWTICO
2013-05-29 4.88      93.13
2013-05-30 4.90      93.57
```

We first plot the raw data for the stock GE, the market-portfolio index $SP500$, and the risk-free interest rate.

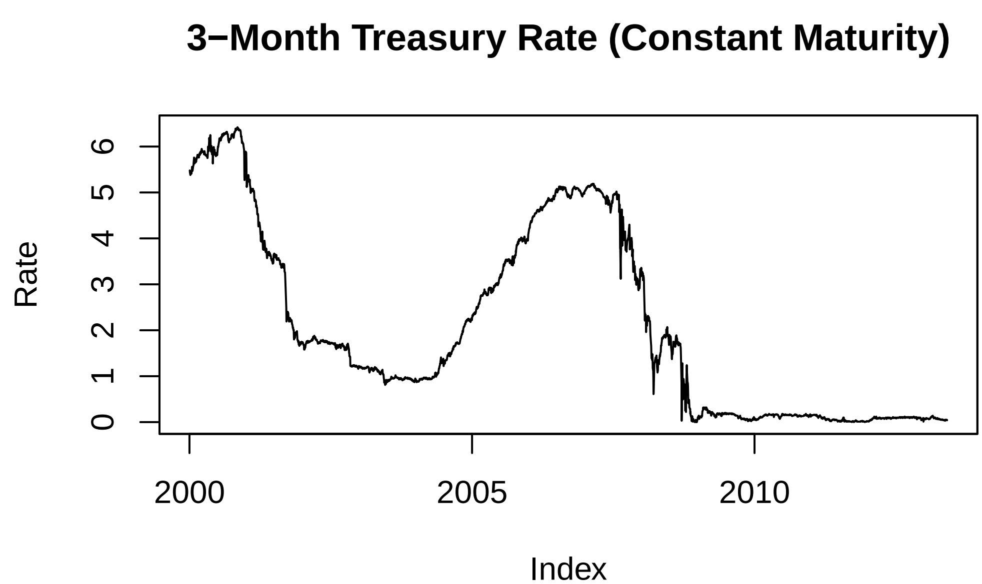
```
plot(casestudy1.data0.00[,"GE"],ylab="Price",main="GE Stock")
```




```
plot(casestudy1.data0.00[,"SP500"], ylab="Value",main="S&P500 Index")
```




```r
plot(casestudy1.data0.00[,"DGS3MO"], ylab="Rate" ,
     main="3-Month Treasury Rate (Constant Maturity)")
```



Now we construct the variables with the log daily returns of GE and the SP500 index as well as the risk-free asset returns.

```
# Compute daily log returns of GE stock
r.daily.GE<-zoo( x=as.matrix(diff(log(casestudy1.data0.00[,"GE"]))),
                 order.by=time(casestudy1.data0.00)[-1])
dimnames(r.daily.GE)[[2]]<-"r.daily.GE"
dim(r.daily.GE)
```

```
[1] 3371    1
```

```
head(r.daily.GE,2)
```

```
             r.daily.GE
2000-01-04 -0.040822037
2000-01-05 -0.001737613
```

```
# Compute daily log returns of the SP500 index
r.daily.SP500<-zoo( x=as.matrix(diff(log(casestudy1.data0.00[,"SP500"]))),
                    order.by=time(casestudy1.data0.00)[-1])
dimnames(r.daily.SP500)[[2]]<-"r.daily.SP500"
dim(r.daily.SP500)
```

```
[1] 3371    1
```

```
head(r.daily.SP500,2)
```

```
           r.daily.SP500
2000-01-04  -0.039099175
2000-01-05   0.001920338
```

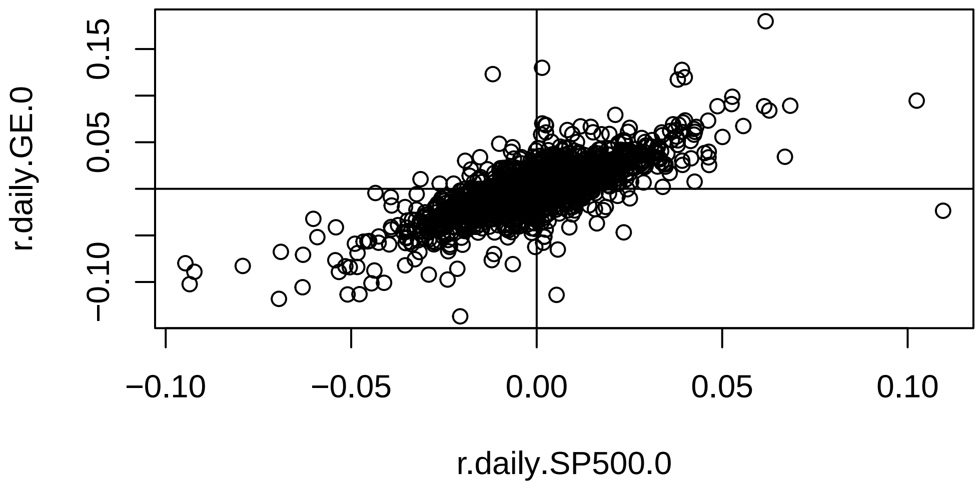
```
# Compute daily return of the risk-free asset
#   accounting for the number of days between successive closing prices
#   apply annual interest rate using 360 days/year
#   (standard on 360-day year since the previous close)

r.daily.riskfree<-log(1 + .01*coredata(casestudy1.data0.00[-1,"DGS3MO"]) *
                        diff(as.numeric(time(casestudy1.data0.00)))/360)
dimnames(r.daily.riskfree)[[2]]<-"r.daily.riskfree"
```

```r
# Compute excess returns (over riskfree rate)
r.daily.GE.0<-r.daily.GE - r.daily.riskfree
dimnames(r.daily.GE.0)[[2]]<-"r.daily.GE.0"

r.daily.SP500.0<-r.daily.SP500 - r.daily.riskfree
dimnames(r.daily.SP500.0)[[2]]<-"r.daily.SP500.0"

# Merge all the time series together,
# and display first and last sets of rows
r.daily.data0<-merge(r.daily.GE, r.daily.SP500, r.daily.riskfree,
                     r.daily.GE.0, r.daily.SP500.0)

head(r.daily.data0,2)
```

```
             r.daily.GE r.daily.SP500 r.daily.riskfree r.daily.GE.0
2000-01-04 -0.040822037  -0.039099175     0.0001508220 -0.040972859
2000-01-05 -0.001737613   0.001920338     0.0001510997 -0.001888713
           r.daily.SP500.0
2000-01-04    -0.039249997
2000-01-05     0.001769238
```

Now we plot the excess returns of GE vs those of the SP500:

```r
plot(r.daily.SP500.0, r.daily.GE.0)
abline(h=0,v=0)
```



7 Fitting the Linear Regression for CAPM

The linear regression model is fit using the R-function `lm( )`:

```
lmfit0 <- lm(r.daily.GE.0 ~ r.daily.SP500.0, data=r.daily.data0)
names(lmfit0) #element names of list object lmfit0
```

```
 [1] "coefficients"  "residuals"     "effects"       "rank"
 [5] "fitted.values" "assign"        "qr"            "df.residual"
 [9] "xlevels"       "call"          "terms"         "model"
```

```
summary.lm(lmfit0) #function summarizing objects created by lm()
```

```

Call:
lm(formula = r.daily.GE.0 ~ r.daily.SP500.0, data = r.daily.data0)

Residuals:
      Min        1Q    Median        3Q       Max
-0.153007 -0.005736 -0.000304  0.005579  0.137347
```

```
Coefficients:
                  Estimate Std. Error t value Pr(>|t|)
(Intercept)     -0.0002523  0.0002385  -1.058     0.29
r.daily.SP500.0  1.1839981  0.0178530  66.319   <2e-16 ***
---
Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

Residual standard error: 0.01384 on 3369 degrees of freedom
Multiple R-squared:  0.5663,    Adjusted R-squared:  0.5661
F-statistic:  4398 on 1 and 3369 DF,  p-value: < 2.2e-16
```

```
lmfit0.summary <- summary(lmfit0)
tstat.intercept <- round(
  lmfit0.summary$coefficients["(Intercept)", "t value"],
  digits=4
  )
```

Note that the t-statistic for the intercept α_{GE} is not significant:

```
print(tstat.intercept)
```

```
[1] -1.0581
```

8 Regression Diagnostics

Some useful R functions

- `anova()`: conduct an Analysis of Variance for the linear regression model, detailing the computation of the F-statistic for no regression structure.
- `influence.measures()`: compute regression diagnostics evaluating case influence for the linear regression model; includes 'hat' matrix, case-deletion statistics for the regression coefficients and for the residual standard deviation.

```
# Compute influence measures (case-deletion statistics)
lmfit0.inflm<-influence.measures(lmfit0)
names(lmfit0.inflm)
```

```
[1] "infmat" "is.inf" "call"
```

```r
dim(lmfit0.inflm$infmat)
```

```
[1] 3371    6
```

```r
head(lmfit0.inflm$infmat,2)
```

```
                 dfb.1_     dfb.r..S        dffit    cov.r       cook.d
2000-01-04  0.007102022 -0.021058608  0.022247223 1.003353 2.475302e-04
2000-01-05 -0.004644642 -0.000630932 -0.004685175 1.000853 1.097846e-05
                    hat
2000-01-04 0.0028523645
2000-01-05 0.0003021269
```

```r
head(lmfit0.inflm$is.inf,2)
```

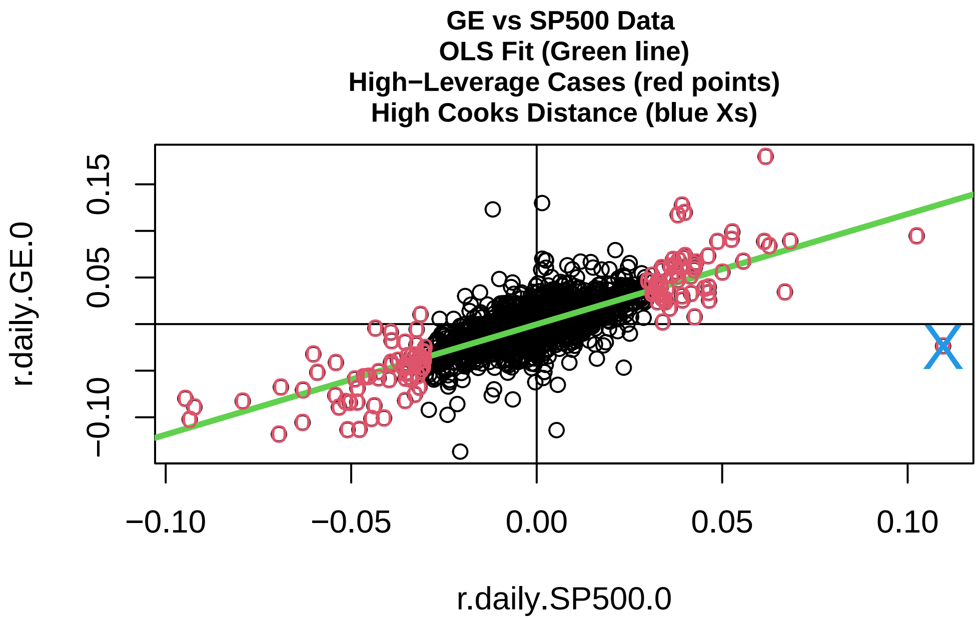
```
           dfb.1_ dfb.r..S dffit cov.r cook.d   hat
2000-01-04  FALSE    FALSE FALSE  TRUE  FALSE  TRUE
2000-01-05  FALSE    FALSE FALSE FALSE  FALSE FALSE
```

```r
# Table counts of influential/non-influential cases
# as measured by the hat/leverage statistic.
table(lmfit0.inflm$is.inf[,"hat"])
```

```
FALSE  TRUE
 3242   129
```

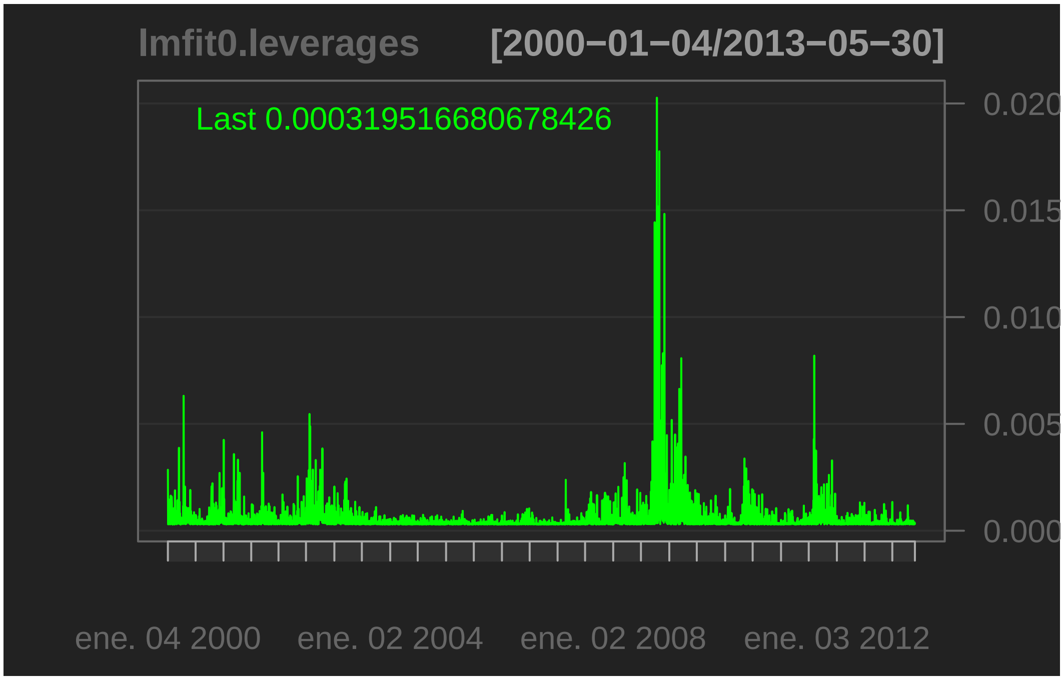
```r
# Re-Plot data adding
#   fitted regression line
#   selective highlighting of influential cases

title_graph <- paste0(
  "GE vs SP500 Data \n",
  "OLS Fit (Green line)\n",
  "High-Leverage Cases (red points)\n",
  "High Cooks Distance (blue Xs)"
  )

plot(
```

```
  r.daily.SP500.0,
  r.daily.GE.0,
  main=title_graph,
  cex.main=0.8
  )
abline(h=0,v=0)
abline(lmfit0, col=3, lwd=3)

# Plot cases with high leverage as red (col=2) "o"s
index.inf.hat<-which(lmfit0.inflm$is.inf[,"hat"]==TRUE)
points(r.daily.SP500.0[index.inf.hat],
       r.daily.GE.0[index.inf.hat],
        col=2, pch="o")

# Plot cases with high cooks distance as big (cex=2) blue (col=4) "X"s
index.inf.cook.d<-which(lmfit0.inflm$is.inf[,"cook.d"]==TRUE)
points(r.daily.SP500.0[index.inf.cook.d],
       r.daily.GE.0[index.inf.cook.d],
       col=4, pch="X", cex=2.)
```




```
# Plot leverage of cases (diagonals of hat matrix)
lmfit0.leverages<-zoo(lmfit0.inflm$infmat[,"hat"],
```

```
                         order.by=time(r.daily.SP500.0))
chartSeries(lmfit0.leverages)
```

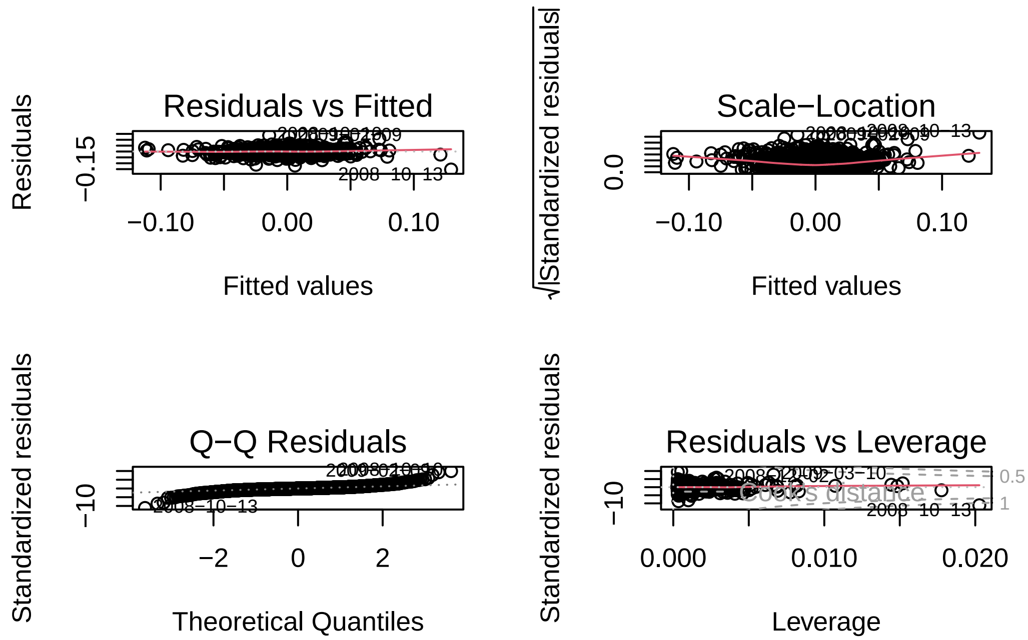


Note the cases are time points of the time series data:

The financial crisis of 2008 is evident.

The R function `plot.lm()` generates a useful 2x2 display of plots for various regression diagnostic statistics:

```
layout(matrix(c(1,2,3,4),2,2)) # optional 4 graphs/page
plot(lmfit0)
```



9 Adding Macro-economic Factors to CAPM

The CAPM relates a stock's return to that of the diversified market portfolio, proxied here by the S&P 500 Index. A stock's return can depend on macro-economic factors, such commodity prices, interest rates, economic growth (GDP).

```
# This section sets up the data frame

# Compute daily log returns of GE stock
r.daily.GE<-zoo( x=as.matrix(diff(log(casestudy1.data0.00[,"GE"]))),
                 order.by=time(casestudy1.data0.00)[-1])
dimnames(r.daily.GE)[[2]]<-"r.daily.GE"

# Repeat for stocks BAC, JDSU, XOM and for commodity DCOILWTICO

# Compute daily log returns of BAC stock
r.daily.BAC<-zoo( x=as.matrix(diff(log(casestudy1.data0.00[,"BAC"]))),
                  order.by=time(casestudy1.data0.00)[-1])
dimnames(r.daily.BAC)[[2]]<-"r.daily.BAC"

# Compute daily log returns of JDSU stock
```

```
r.daily.JDSU<-zoo( x=as.matrix(diff(log(casestudy1.data0.00[,"JDSU"]))),
                   order.by=time(casestudy1.data0.00)[-1])
dimnames(r.daily.JDSU)[[2]]<-"r.daily.JDSU"

# Compute daily log returns of XOM stock
r.daily.XOM<-zoo( x=as.matrix(diff(log(casestudy1.data0.00[,"XOM"]))),
                  order.by=time(casestudy1.data0.00)[-1])
dimnames(r.daily.XOM)[[2]]<-"r.daily.XOM"

# Compute daily log returns of DCOILWTICO (Crude Oil WTI)
r.daily.DCOILWTICO<-zoo(
  x=as.matrix(diff(log(casestudy1.data0.00[,"DCOILWTICO"]))),
  order.by=time(casestudy1.data0.00)[-1])
dimnames(r.daily.DCOILWTICO)[[2]]<-"r.daily.DCOILWTICO"

# Compute daily log returns of the SP500 index
r.daily.SP500<-zoo( x=as.matrix(diff(log(casestudy1.data0.00[,"SP500"]))),
                    order.by=time(casestudy1.data0.00)[-1])
dimnames(r.daily.SP500)[[2]]<-"r.daily.SP500"


# Compute daily log change interest rate variables
dlog.daily.DGS1<-zoo( x=as.matrix(diff(log(casestudy1.data0.00[,"DGS1"]))),
                      order.by=time(casestudy1.data0.00)[-1])
dimnames(dlog.daily.DGS1)[[2]]<-"dlog.daily.DGS1"

dlog.daily.DGS10<-zoo( x=as.matrix(diff(log(casestudy1.data0.00[,"DGS10"]))),
                      order.by=time(casestudy1.data0.00)[-1])
dimnames(dlog.daily.DGS10)[[2]]<-"dlog.daily.DGS10"

dlog.daily.DAAA<-zoo( x=as.matrix(diff(log(casestudy1.data0.00[,"DAAA"]))),
                       order.by=time(casestudy1.data0.00)[-1])
dimnames(dlog.daily.DAAA)[[2]]<-"dlog.daily.DAAA"

dlog.daily.DBAA<-zoo( x=as.matrix(diff(log(casestudy1.data0.00[,"DBAA"]))),
                      order.by=time(casestudy1.data0.00)[-1])
dimnames(dlog.daily.DBAA)[[2]]<-"dlog.daily.DBAA"

# Compute daily return of the risk-free asset
# (accounting for the number of days since the previous close)

r.daily.riskfree<-log(1 + .01*coredata(casestudy1.data0.00[-1,"DGS3MO"]) *
```

```
                        diff(as.numeric(time(casestudy1.data0.00)))/360)
dimnames(r.daily.riskfree)[[2]]<-"r.daily.riskfree"

# Compute excess returns (over riskfree rate)

r.daily.GE.0<-r.daily.GE - r.daily.riskfree
  dimnames(r.daily.GE.0)[[2]]<-"r.daily.GE.0"

r.daily.BAC.0<-r.daily.BAC - r.daily.riskfree
dimnames(r.daily.BAC.0)[[2]]<-"r.daily.BAC.0"

r.daily.JDSU.0<-r.daily.JDSU - r.daily.riskfree
dimnames(r.daily.JDSU.0)[[2]]<-"r.daily.JDSU.0"

r.daily.XOM.0<-r.daily.XOM - r.daily.riskfree
dimnames(r.daily.XOM.0)[[2]]<-"r.daily.XOM.0"

r.daily.SP500.0<-r.daily.SP500 - r.daily.riskfree
  dimnames(r.daily.SP500.0)[[2]]<-"r.daily.SP500.0"

r.daily.DCOILWTICO.0<-r.daily.DCOILWTICO - r.daily.riskfree
dimnames(r.daily.DCOILWTICO.0)[[2]]<-"r.daily.DCOILWTICO.0"

# Merge all the time series together, and display first and last sets of rows
r.daily.data0<-merge(r.daily.GE, r.daily.SP500, r.daily.riskfree,
                     r.daily.GE.0, r.daily.SP500.0)

r.daily.data00<-merge(
  r.daily.GE, r.daily.SP500, r.daily.riskfree,
  r.daily.GE.0, r.daily.SP500.0, r.daily.DCOILWTICO.0,
  r.daily.BAC, r.daily.JDSU, r.daily.XOM,
  r.daily.BAC.0, r.daily.JDSU.0, r.daily.XOM.0,
  dlog.daily.DGS1, dlog.daily.DGS10,
  dlog.daily.DAAA, dlog.daily.DBAA)

print(names(r.daily.data00))
```

```
 [1] "r.daily.GE"           "r.daily.SP500"        "r.daily.riskfree"
 [4] "r.daily.GE.0"         "r.daily.SP500.0"      "r.daily.DCOILWTICO.0"
 [7] "r.daily.BAC"          "r.daily.JDSU"         "r.daily.XOM"
[10] "r.daily.BAC.0"        "r.daily.JDSU.0"       "r.daily.XOM.0"
[13] "dlog.daily.DGS1"      "dlog.daily.DGS10"     "dlog.daily.DAAA"
```

```
[16] "dlog.daily.DBAA"
```

```r
# The linear regression for the simple CAPM:
lmfit0<-lm(r.daily.GE.0 ~ r.daily.SP500.0,
           data=r.daily.data00)
summary.lm(lmfit0)
```

```
Call:
lm(formula = r.daily.GE.0 ~ r.daily.SP500.0, data = r.daily.data00)

Residuals:
      Min        1Q    Median        3Q       Max 
-0.153007 -0.005736 -0.000304  0.005579  0.137347 

Coefficients:
                  Estimate Std. Error t value Pr(>|t|)    
(Intercept)     -0.0002523  0.0002385  -1.058     0.29    
r.daily.SP500.0  1.1839981  0.0178530  66.319   <2e-16 ***
---
Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

Residual standard error: 0.01384 on 3369 degrees of freedom
Multiple R-squared:  0.5663,    Adjusted R-squared:  0.5661 
F-statistic:  4398 on 1 and 3369 DF,  p-value: < 2.2e-16
```

```r
# The linear regression for the extended CAPM:
lmfit1<-lm(r.daily.GE.0 ~ r.daily.SP500.0 + r.daily.DCOILWTICO,
           data=r.daily.data00)
summary.lm(lmfit1)
```

```
Call:
lm(formula = r.daily.GE.0 ~ r.daily.SP500.0 + r.daily.DCOILWTICO, 
    data = r.daily.data00)

Residuals:
      Min        1Q    Median        3Q       Max 
-0.152804 -0.005563 -0.000254  0.005654  0.133477 

Coefficients:
```

```
                     Estimate Std. Error t value Pr(>|t|)
(Intercept)        -0.0002396  0.0002381  -1.006  0.31438
r.daily.SP500.0     1.1976245  0.0181871  65.850  < 2e-16 ***
r.daily.DCOILWTICO -0.0363206  0.0096484  -3.764  0.00017 ***
---
Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

Residual standard error: 0.01382 on 3367 degrees of freedom
  (1 observation deleted due to missingness)
Multiple R-squared:  0.5676,    Adjusted R-squared:  0.5674
F-statistic:  2210 on 2 and 3367 DF,  p-value: < 2.2e-16
```

The regression coefficient for the oil factor (*r.daily.DCOILWTICO*) is statistically significant and negative. Over the analysis period, price changes in GE stock are negatively related to the price changes in oil.

10 Analysis of Variance

```r
anova(lmfit1)
```

```
Analysis of Variance Table

Response: r.daily.GE.0
                     Df  Sum Sq Mean Sq  F value    Pr(>F)
r.daily.SP500.0       1 0.84142 0.84142 4405.871 < 2.2e-16 ***
r.daily.DCOILWTICO    1 0.00271 0.00271   14.171 0.0001698 ***
Residuals          3367 0.64302 0.00019
---
Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
```

```r
anova(lmfit0)
```

```
Analysis of Variance Table

Response: r.daily.GE.0
                  Df  Sum Sq Mean Sq F value    Pr(>F)
r.daily.SP500.0    1 0.84304 0.84304  4398.2 < 2.2e-16 ***
Residuals       3369 0.64576 0.00019
---
Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
```

```
#anova(lmfit0, lmfit1)
```

11 Exxon-Mobil stock

Consider the corresponding models for the oil stock XOM (Exxon-Mobil)

```
# The linear regression for the simple CAPM:
lmfit0<-lm(r.daily.XOM.0 ~ r.daily.SP500.0,
           data=r.daily.data00)
summary.lm(lmfit0)
```

```
Call:
lm(formula = r.daily.XOM.0 ~ r.daily.SP500.0, data = r.daily.data00)

Residuals:
      Min        1Q    Median        3Q       Max
-0.085338 -0.005785 -0.000009  0.006233  0.113781

Coefficients:
                 Estimate Std. Error t value Pr(>|t|)
(Intercept)     0.0002079  0.0002108   0.986    0.324
r.daily.SP500.0 0.8315939  0.0157846  52.684   <2e-16 ***
---
Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

Residual standard error: 0.01224 on 3369 degrees of freedom
Multiple R-squared:  0.4517,    Adjusted R-squared:  0.4516
F-statistic:  2776 on 1 and 3369 DF,  p-value: < 2.2e-16
```

```
# The linear regression for the extended CAPM:
lmfit1<-lm(r.daily.XOM.0 ~ r.daily.SP500.0 + r.daily.DCOILWTICO.0,
            data=r.daily.data00)
summary.lm(lmfit1)
```

```
Call:
lm(formula = r.daily.XOM.0 ~ r.daily.SP500.0 + r.daily.DCOILWTICO.0,
    data = r.daily.data00)
```

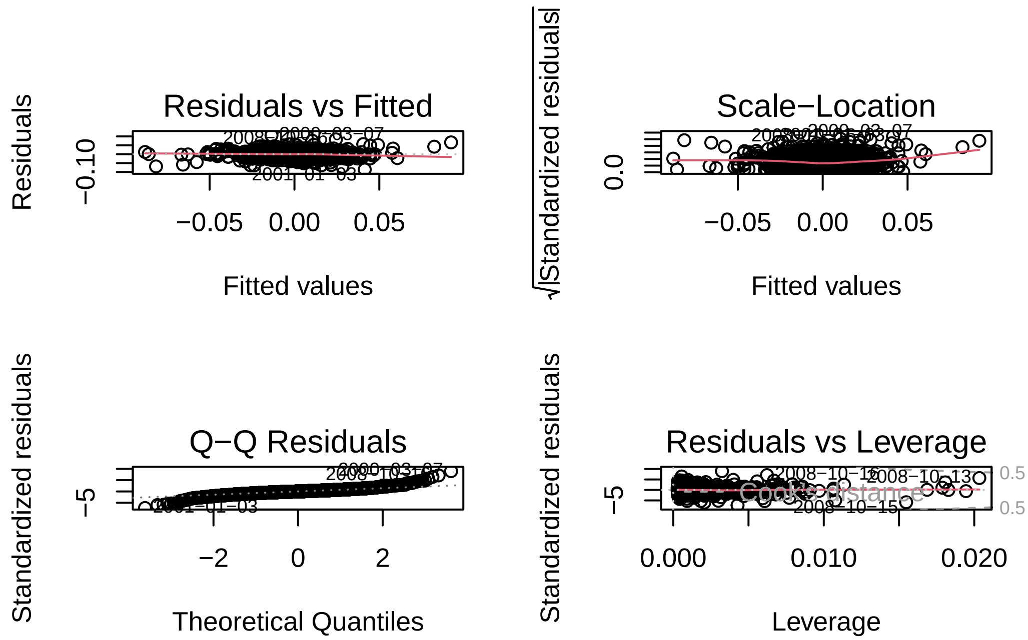
```
Residuals:
      Min        1Q    Median        3Q       Max 
-0.086025 -0.005586 -0.000006  0.005805  0.105784 

Coefficients:
                      Estimate Std. Error t value Pr(>|t|)    
(Intercept)          0.0001627  0.0002033    0.80    0.424    
r.daily.SP500.0      0.7842980  0.0155314   50.50   <2e-16 ***
r.daily.DCOILWTICO.0 0.1319092  0.0082391   16.01   <2e-16 ***
---
Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

Residual standard error: 0.0118 on 3367 degrees of freedom
  (1 observation deleted due to missingness)
Multiple R-squared:  0.4905,    Adjusted R-squared:  0.4902 
F-statistic:  1620 on 2 and 3367 DF,  p-value: < 2.2e-16
```

Results:

- The R-squared for XOM is lower than for GE. Its relationship to the market index is less strong.
- The beta of XOM with the SP500 is less than 1.
- The regression coefficient for the oil factor ($r.daily.DCOILWTICO$) is statistically significant and positive.

For the extended model, we use the R function *plot.lm*() to display regression diagnostic statistics:

```r
layout(matrix(c(1,2,3,4),2,2)) # optional 4 graphs/page
plot(lmfit1)
```



The high-leverage cases in the data are those which have high Mahalanobis distance from the center of the data in terms of the column space of the independent variables (see: Fall 2013 Regression Analysis Problem Set).

We display the data in terms of the independent variables and highlight the high-leverage cases.

```r
# Refit the model using argument x=TRUE so that the lm object includes the
# matrix of independent variables
lmfit1<-lm(r.daily.XOM.0 ~ r.daily.SP500.0 + r.daily.DCOILWTICO,
           data=r.daily.data00,
           x=TRUE)
names(lmfit1)
```

```
 [1] "coefficients"  "residuals"     "effects"       "rank"
 [5] "fitted.values" "assign"        "qr"            "df.residual"
 [9] "na.action"     "xlevels"       "call"          "terms"
[13] "model"         "x"
```

```r
dim(lmfit1$x)
```

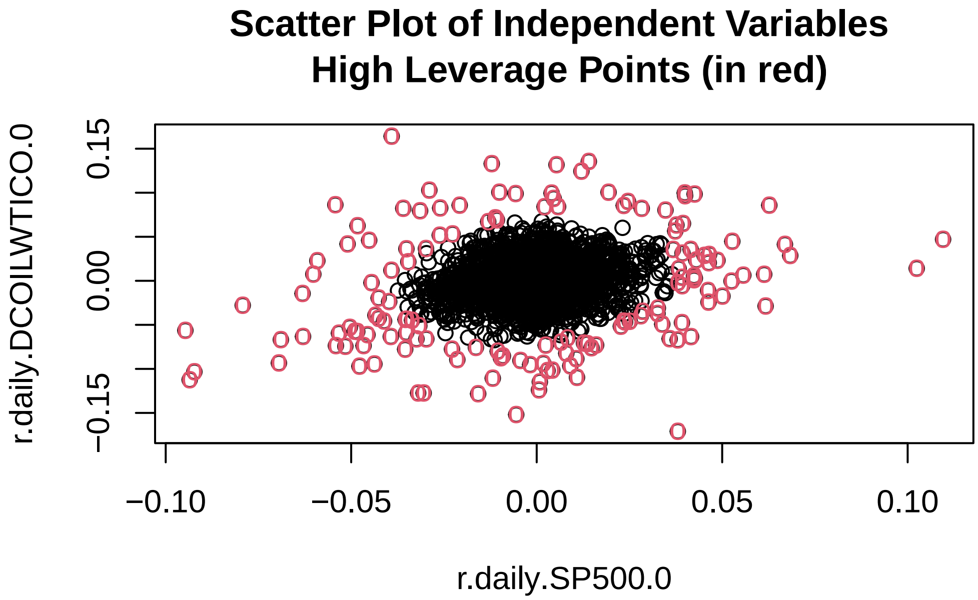
```
[1] 3370    3
```

```r
head(lmfit1$x)
```

```
           (Intercept) r.daily.SP500.0 r.daily.DCOILWTICO
2000-01-05           1    0.0017692383       -0.036251729
2000-01-06           1    0.0008049553        0.005663446
2000-01-07           1    0.0265805165        0.000000000
2000-01-10           1    0.0106762602       -0.003232326
2000-01-11           1   -0.0132993987        0.038893791
2000-01-12           1   -0.0045473971        0.023467128
```

We now compute the leverage (and other influence measures) with the function *influence.measures*() and display the scatter plot of the independent variables, highlighting the high-leverage cases.

```r
lmfit1.inflm<-influence.measures(lmfit1)
index.inf.hat<-which(lmfit1.inflm$is.inf[,"hat"]==TRUE)

par(mfcol=c(1,1))
plot(lmfit1$x[,2],
     lmfit1$x[,3],
     xlab="r.daily.SP500.0",
     ylab="r.daily.DCOILWTICO.0")
title(
  main="Scatter Plot of Independent Variables \n High Leverage Points (in red)"
  )

points(lmfit1$x[index.inf.hat,2],
       lmfit1$x[index.inf.hat,3],
       col=2,
       pch="o")
```


Scatter Plot of Independent Variables
High Leverage Points (in red)
r.daily.DCOILWTICO.0
−0.15
0.00
0.15
−0.10
−0.05
0.00
0.05
0.10
r.daily.SP500.0